

1. Diamond Princess

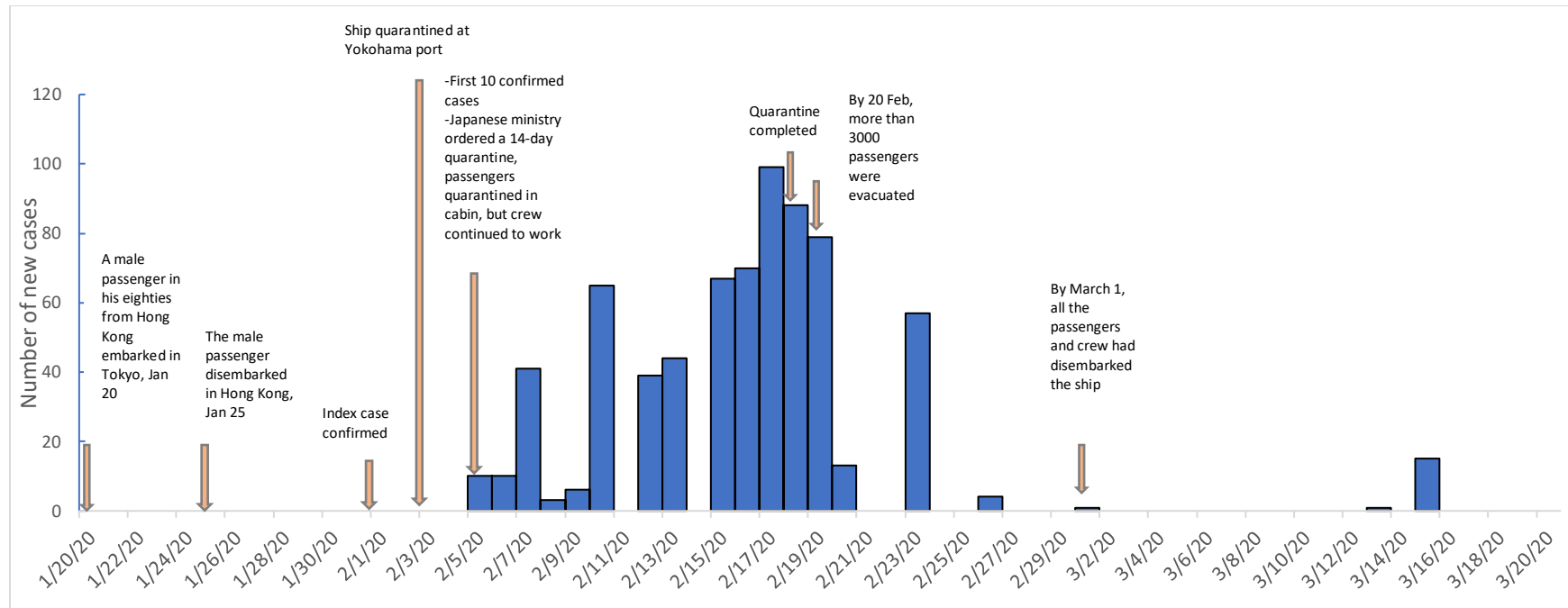


Figure 1. New COVID-19 positive cases in Diamond Princess

On January 20, 2020 Diamond Princess sailed from Yokohama, Japan with 3711 passengers and crew (2666 passengers and 1045 crew) for a 14-day cruise to China, Vietnam, and Taiwan and then back to Japan [1] [2]. On 25th January, a male passenger in his eighties from Hongkong who became passenger in Tokyo disembarked in Hong Kong which was confirmed to be coronavirus positive on February 1 [3] [4]. With the notification of a SARS-Cov-2 positive in the passenger, the ship was quarantined at Yokohama port on 3rd February. Passengers were informed about social distancing and symptom monitoring [2]. On 5th February, first 10 confirmed cases of COVID-19 were reported from the ship [3]. All the confirmed cases including ill passengers/crew were taken to Japanese hospitals while 14-day quarantine order was issued for all other passengers and crew on board by the Japanese ministry [5]. After quarantine began, passengers required to be in their cabins and were allowed to come out of cabin for 1 hours in a day. The symptomatic crew were also required to be in their cabins while the non-symptomatic crew continued to perform their regular duties including delivering the meals[6]. By February 11, Japanese government had tested 439 of 3711 people on board, with 135 people testing positive which was a higher number than the total confirmed cases of Japan at that time [7]. On

February 17, the quarantine period completed, and by 20th February, more than 3000 passengers were evacuated using mostly the air-evacuation [3]. By March 1, all the passengers and crew had disembarked the ship. A total of 712 cases (567 passengers, and 145 crew) were confirmed as coronavirus cases by March 15 and a total of 13 people were reported to have died due to COVID-19 by April 16, 2020 [8].

2. Grand Princess

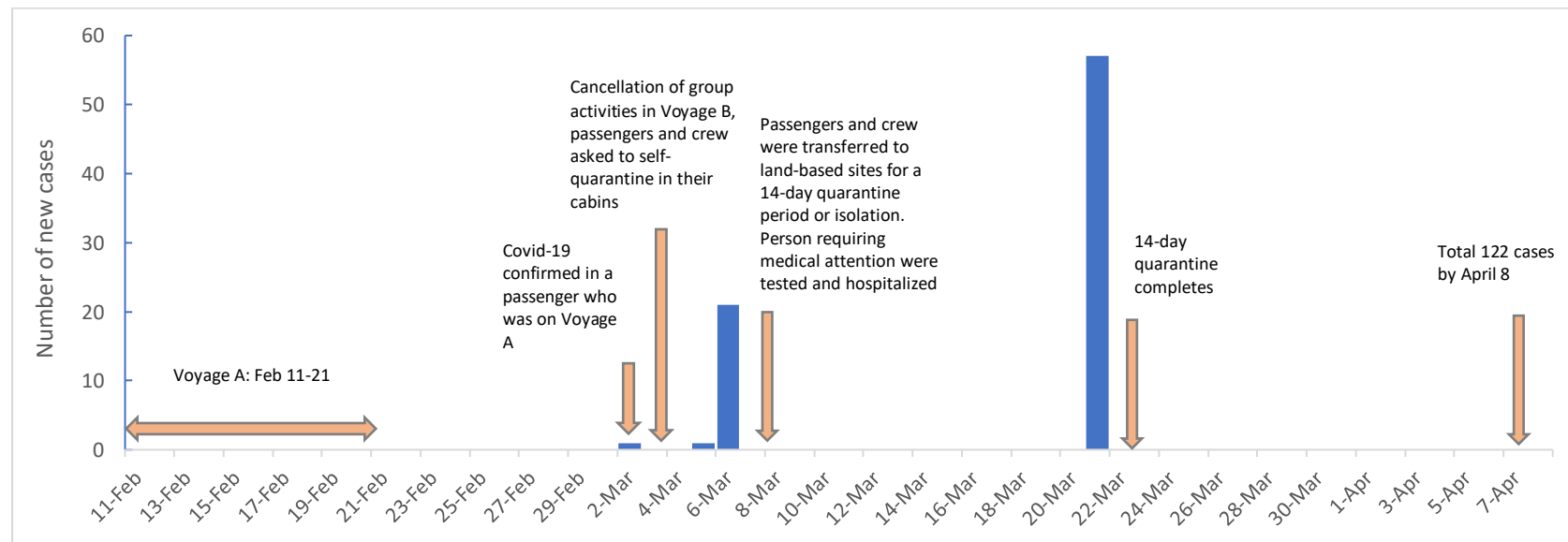


Figure 2: Grand Princess new COVID-19 cases and other events in the timeline

Voyage A of Grand Princess started with 1179 passengers and crew (1111 crew and 68 passengers) from San Francisco to Mexico from February 11-21. Voyage B started on February 21 with 3571 people (1111 crew and 2460 passengers) including most of the passengers from Voyage A from San Francisco to Hawaiian Islands[2]. On March 2, coronavirus was confirmed in a person who was on Voyage A [9]. On March 4, the first confirmed case died and from the same day onwards group activities were cancelled in Voyage B, passengers and symptomatic

crew members were asked to self-quarantine in their cabins, public dining were replaced by room service until disembarkation [2]. Subsequently on March 5, another man on voyage A was confirmed with coronavirus [9] followed by 21 new confirmed cases in Voyage B on March 6. On March 8, passengers and crew were transferred to land-based sites for a 14-day quarantine period[2]. All the person requiring medical attention were tested and hospitalized[2]. After 14 day of quarantine, Grand Princess passengers were allowed to leave on 23rd March. Crew members on board completed quarantine on April 4. By April 8, a total of 5 deaths and 122 cases (103 passengers and 19 crew) confirmed as SARS-COV-2 positive were reported [10] [11].

3. Ruby Princess

On February 19, 2020 New South Wales (NSW) developed a draft guideline for cruise ships that the ships required to update a full acute disease log, 24 hour before arriving in the port and that if respiratory outbreak affected 1% of ships passengers, the ship would be labelled as medium to high risk. For the ships with medium to high risk, a NSW health assessment team must board the ship prior to the disembarkation of the passengers [12]. Ruby Princess that left Sydney for a 13-day cruise to New Zealand on 24th February 2020 was scheduled to return to Sydney on 8th March. In line with the NSW guideline, it reported to NSW on March 7 that there were 30 passengers on board who had influenza-like-illness (ILI), and two of them were from countries affected by Covid-19 which increased to 170 in the log on March 8 [12]. On March 8, before disembarking the passengers, NSW health assessment team reported 360 passengers and crew with respiratory symptoms of whom 9 swabs were taken for Covid-19 test[12]. Despite the larger number of people with respiratory symptom, the ship was allowed to disembark the passengers from February 24 trip, and then embarked 2700 passengers for a second trip to New Zealand. The ship departed on March 8 with 3800 people (2700 passengers and 1100 crew) [13]. On March 14, the ship docked in Wellington and 5 Covid-19 test swabs were taken from the sick passengers who resulted negative later [12]. On March 15, the ship docked on Napier and passengers were still enjoying onshore. On March 17, 104 people on board showed acute respiratory symptoms of whom 26 had ILI. Then on March 19, the ship arrived back in Sydney where 2 sick passengers were taken to hospital and all other passengers were allowed to disembark. Throughout the journey, passengers were not told to practice social distancing and were not provided any information on Covid-19. Upon disembarkation, the passengers took different routes such as train, bus, and international flights. Then on March 20, 3 passengers and 1 crew member were reported to have tested positive for coronavirus [13]. The number of confirmed cases rose to 48 by March 22, 133 by March 24, and 521 (140 crew on board, 12 crew in health facilities, and 369 passengers) by April 15. By May 9, a total of about 700 cases and 21 deaths due to covid-19 were reported to be associated with Ruby Princess passengers and crew.

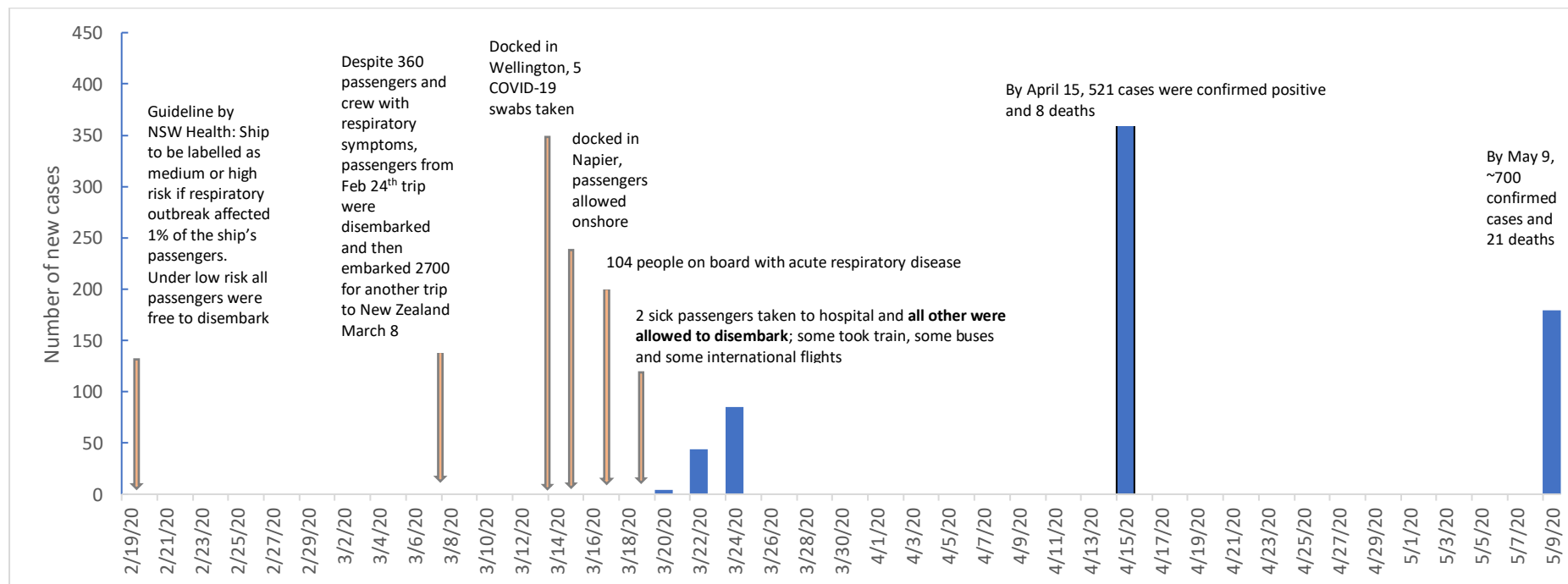


Figure 3: COVID-19 new cases and event in timeline in Ruby Princess

4. World Dream

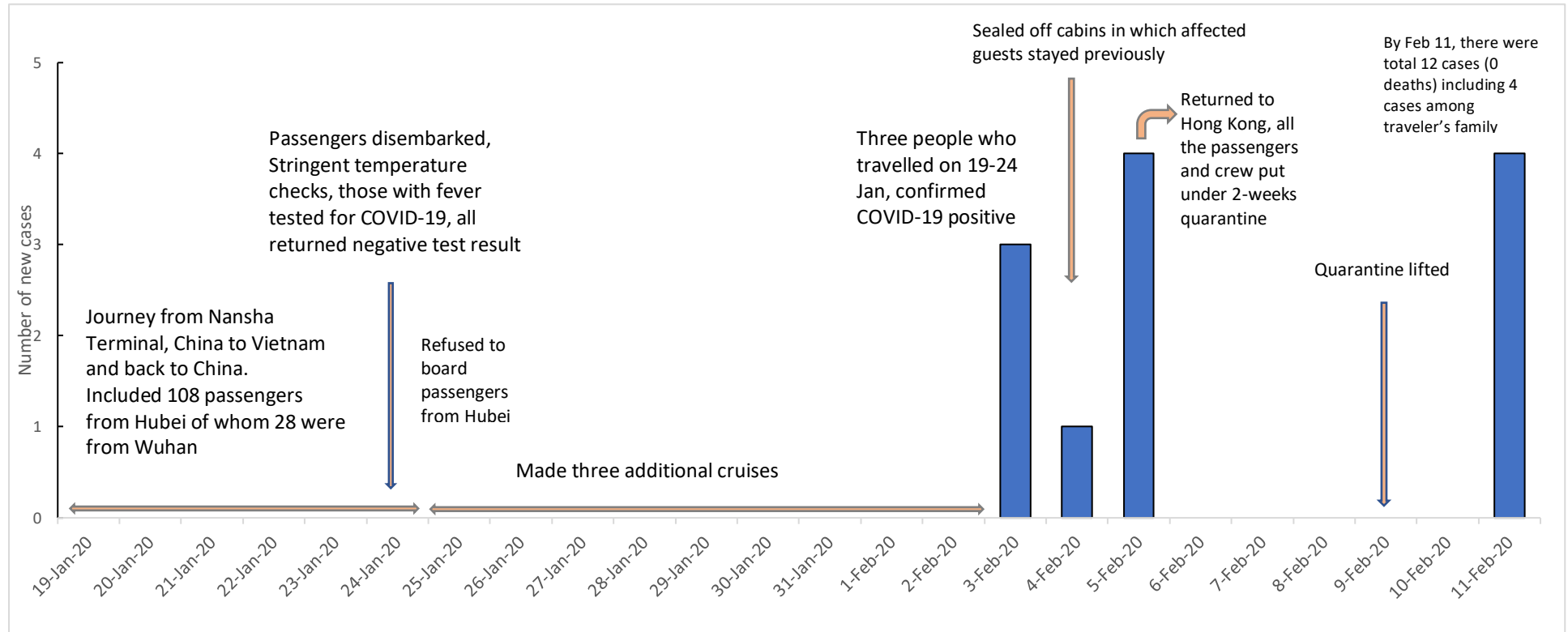


Figure 4: COVID-19 new cases in World Dream Cruise ship

World dream started journey from Nansha Terminal, China to Vietnam and back to China from 19-24 January, 2020 which included 108 passengers from Hubei of whom 28 were from Wuhan. On January 24, upon disembarkation all the passengers were checked for temperature in Guangzhou and those with fever were further tested for coronavirus by the Nansha Center for Disease Control and Prevention. The test

result was negative for the total test at that time [14]. On January 24, passengers from Hubei province were refused to board the ship. On 3rd February, 3 people who travelled in the ship from 19-24 January were confirmed as SARS COV 2 positive. So, on February 4, Taiwan did not allow disembarkation in Kaohsiung port and the ship was returned to Hong Kong and also the ship sealed off the cabins in which the infected passengers stayed earlier. On the same day on February 4, a temporary part-time ground worker on the "World Dream" cruise ship was confirmed to be SARS-Cov-2 positive [15]. On February 5, the number of confirmed cases among passengers in 19-24 Jan voyage increased to 8. On February 5, all the passengers and crew were put under 2-weeks quarantine in Hong Kong that was lifted in Feb 9 when all the crew (1800) resulted in test negative. Majority of the passengers were not tested as they had had no contact with the infected passengers on board during 19-24 January [16]. By February 11, there were total 8 confirmed cases among the crew and passengers, and 12 cases including their 4 family members. No any deaths were reported[6].

5. Asara

On February 1, 2020 a Taiwanese American who travelled on Asara cruise ship, also called as MS River Anuket left Egypt. On March 1, Egyptian health authorities were notified that the Taiwanese American woman had been carrying the virus while on the Asara and may have infected others on the ship. And so the samples of crew were sent for Covid-19 test [17]. On March 5, the ship sailed for trip from Aswan to Luxor with 171 people on board (101 passenger and 70 crew). On March 6, the test results confirmed 12 cases of coronavirus among the crew, all of whom were symptomatic. On the same day, in March 6, 3 residents of Montgomery county, Md tested positive for the virus (asymptomatic) who were found to have contracted the virus while they were on Asara cruise. The three cases arrived US on Feb 20 and were tested for the virus on March 4. The case of Asara reflects, how asymptomatic cases can transmit the disease in the community. By the time the cases were confirmed, one attended an event near Philadelphia, came in contact with students and teachers from a local school (5 schools were closed in that district later). Likewise, the second case had went to a retirement community in Rockville, Md and came in contact with 70-100 people. On March 7, 33 new cases (making a total of 45 cases) were confirmed in Asara cruise ship who were sent to hospital and the remaining passengers and crew were quarantined on the ship which was lifted on March 21. It has been reported that atleast 21 American who returned to the US who were apparently in Asara from late Feb to early March were confirmed to have virus [18].

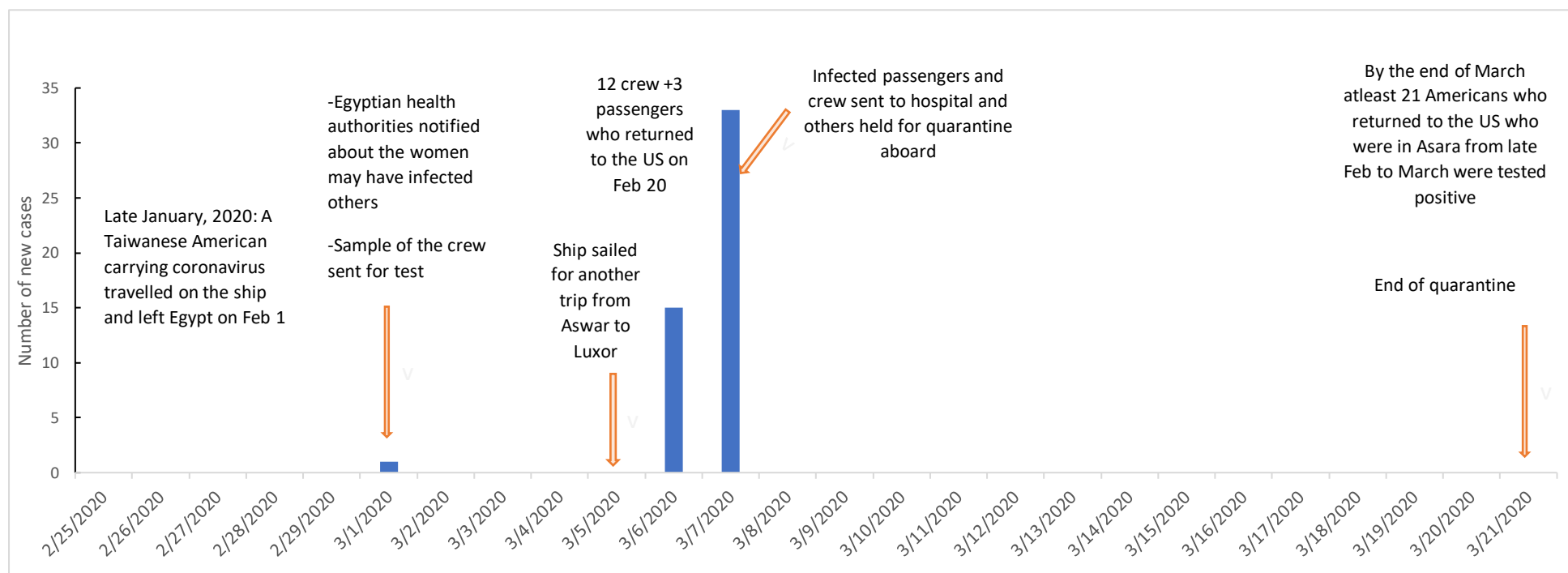


Figure 5: COVID-19 new cases in Asara cruise ship

6. Costa Luminosa

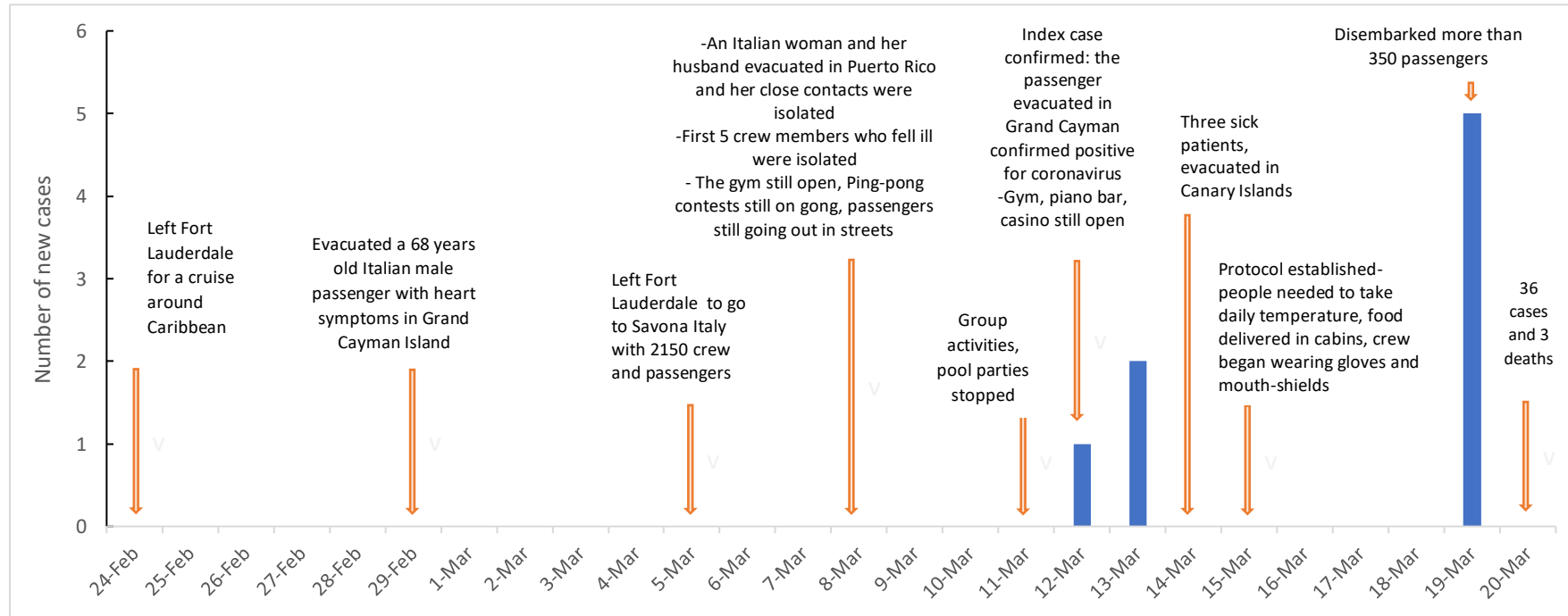


Figure 6: COVID-19 new cases and events in Costa Luminosa cruise ship

Costa Luminosa, first left Fort Lauderdale for a cruise around Caribbean on Feb 24. As coronavirus was taking tolls in Italy, Jamaica did not allow to disembark Italian passengers as scheduled on Feb 28. So, it evacuated a 69 year old Italian male passenger with heart symptoms in Grand Cayman Island on Feb 29 and then returned to Fort Lauderdale to drop some passenger and pick more [19]. Then it left Fort Lauderdale on March 5 destined to go to Venice, Italy to Puerto Rico to Tenerife to Marseille, France to Savona, Italy [20]. On March 7, a 68 year old Italian woman showed worsening respiratory condition who along with her husband were evacuated in Puerto Rico and her close contacts were

isolated[19] on March 8. On the same day, first five crew members who fell ill were isolated [21].However, other passengers uninformed of their risk were still going out and enjoying their day, the gym stayed open and the Ping-pong contests were still ongoing [19]. On March 9, the man left in Cayman Island on late February started developing dry cough and was then tested for coronavirus [19]. On March 11, the cruise stopped group activities like lounge and pool parties[19] while the casino, piano bar, karaoke, gym still stayed open until March 12. On the same day on March 12, first Covid-19 case was confirmed on the man who was tested for Covid-19 on March 8[22]. On the other side, the woman and her husband who were evacuated in Puerto Rico were confirmed test positive on March 13 [19] . Only after then on March 15, protocol for an outbreak on board was established, people were to take their temperature every day, all 1421 passengers at that time were isolated to their rooms, food was delivered on their cabins, crew began wearing gloves, gowns and mouth shields (table napkins), people with inside rooms were moved to the cabins with balconies [19]. On March 19, the ship disembarked more than 350 passengers (235 Americans, and 77 Canadians[23]), at Marseille, France. While in flight, five of the tested passengers returned positive [24], so those whose results were positive were sat in separated seats [25]. Similarly, 10 Spanish and 187 French nationals were also disembarked [23]. All the passengers who disembarked were given face masks and were asked to wear them until they would be confined in their homes [23]. By March 20, a total of 36 cases and 3 deaths were reported [23, 26].

7. Artania

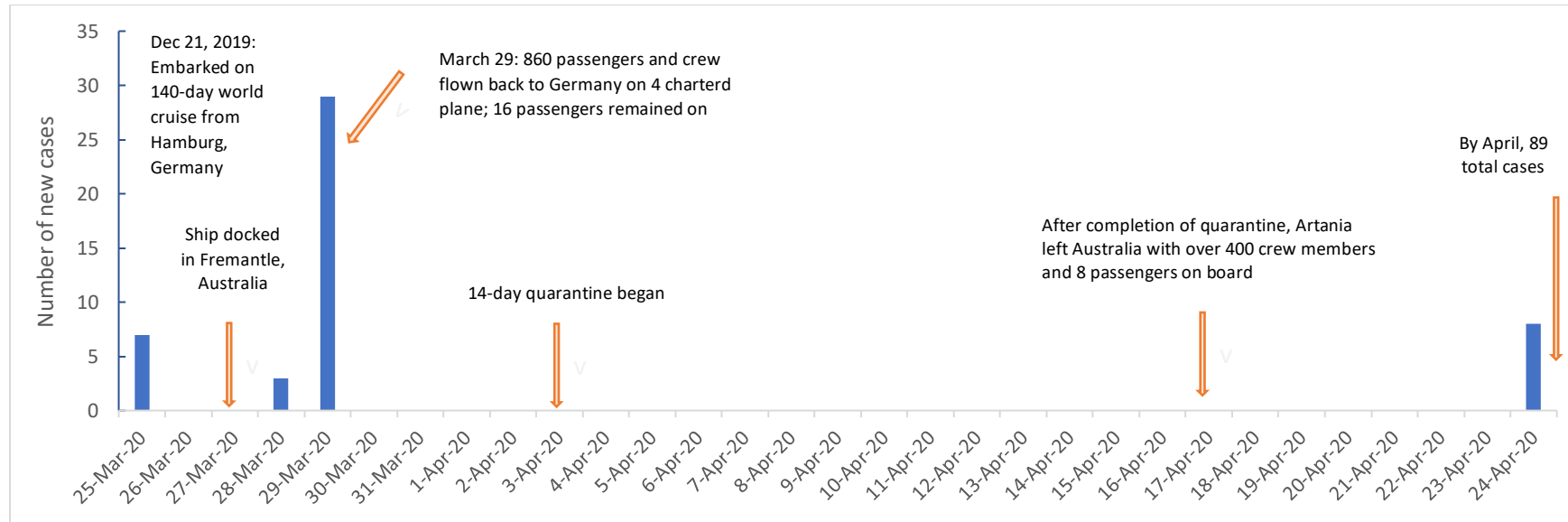


Figure 7: COVID-19 new cases in Artania cruise ship

On December 21, 2019 the Artania cruise ship embarked on a 140-day world cruise from Hamburg, Germany, to Bremerhaven, Germany. On March 25, ship stopped in Fremantle for refuel and it was also the time when people on board began showing covid-19 symptoms. When tested, 7 of 9 tests showed positive and they were taken to hospital [27]. On March 27, the ship docked in Fremantle in Western Australia (WA) and a day later on March 28, there were 3 new confirmed cases [28]. The following day on 29th March, 29 cases were confirmed as positive among 46 tests done in crew and passengers [28]. On March 29, different repatriation flights took 860 passengers and crew back to Germany while 12-16 passengers including those who were too ill to fly remained in board [27]. By late March at least 36 people on board has tested positive for coronavirus. The remaining crew and passengers were kept in quarantine starting on April 3 [27]. After the completion of quarantine period on April 17, Artania cruise ship left Australia with over 400 crew members on board [27]. On April 24, 8 crew were reported to have tested positive. By the end of April, 4 deaths were reported and Artania was held responsible for 81 of the Western Australia's total cases.

8. Zaandam and Rotterdam

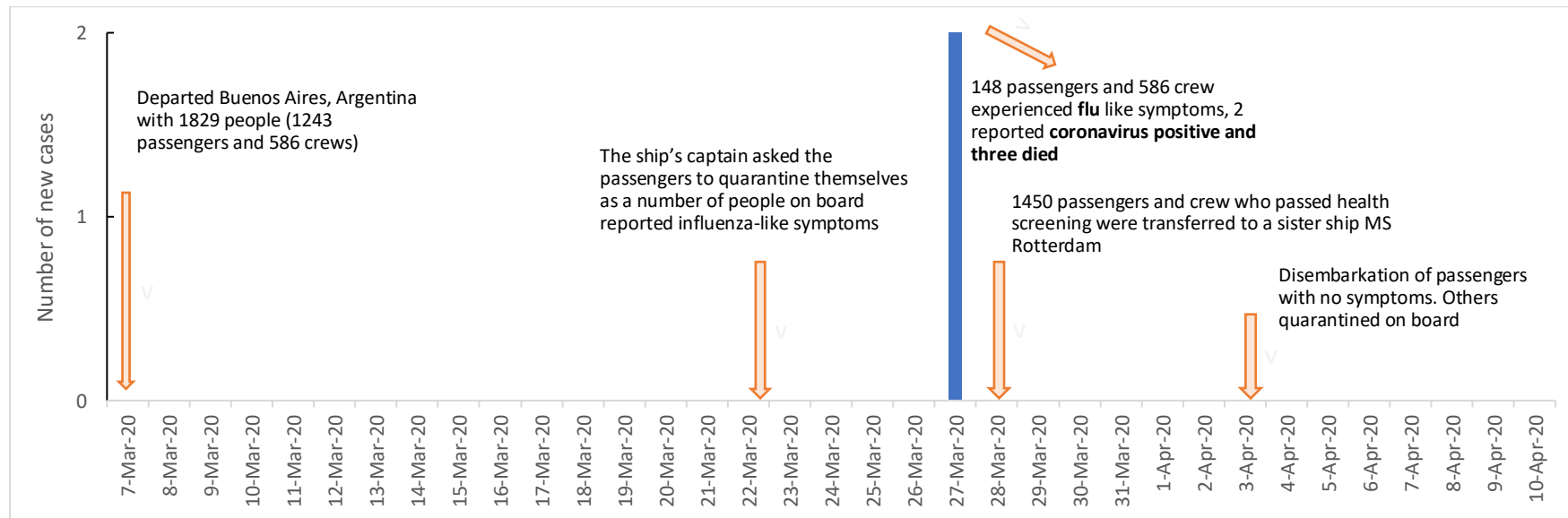


Figure 8. Timeline of events in Zaandam cruise ship

On March 7, 2020 Zaandam departed Buenos Aires with its 1243 passengers and 586 crew. On March 27 as a number of people on board reported influenza like symptoms, the ship's captain asked the passengers to quarantine themselves [29]. On March 21, Chile did not allow the ship to dock. On March 27, 148 passengers and 586 crew experienced flu like symptoms, 2 reported positive and three died[30]. On the same day it was denied transit to Panama Canal. On March 28, 1450 passengers and crew who passed health screening (healthy patient who showed no symptoms, temperature screening and questionnaire) were transferred to a sister ship MS Rotterdam to decrease the load to crew as many were getting ill. During the transfer, social distancing was practiced [29]. The ship was allowed to dock in Florida on April 1. On April 2, a total of 107 passengers (90 on Zaandam, and 19 on Rotterdam) and 143 crew presented with influenza like symptoms[31]. On the following day, on April 3, passengers with no symptoms were disembarked. Those with symptoms stayed onboard until they fully recovered and were fit to travel as per CDC guideline[31]. By April 10, a total of 8 cases and 4 deaths were reported.

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